PROGRESS LOG

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Circuit building and testing

Printing phrases

First, we tried printing phrases of words on the LCD display.
Connections made:

Example code for printing phrases:

```
#include <LiquidCrystal.h>

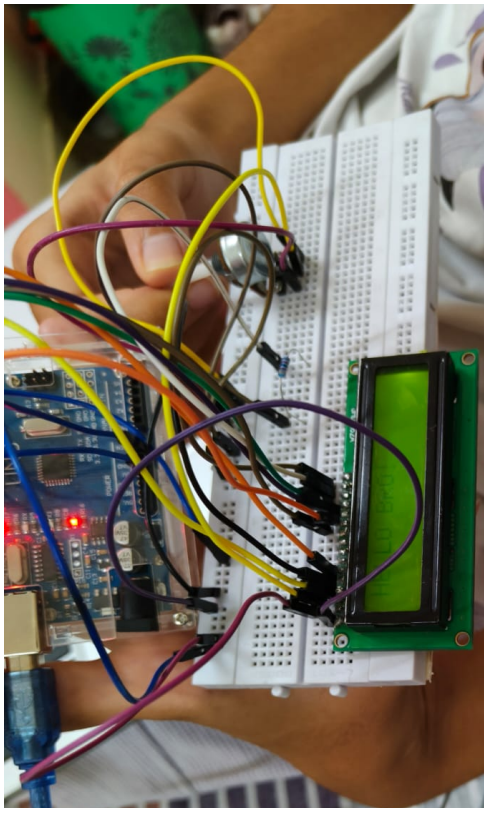
// RS, E, DB4, DB5, DB6, DB7
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

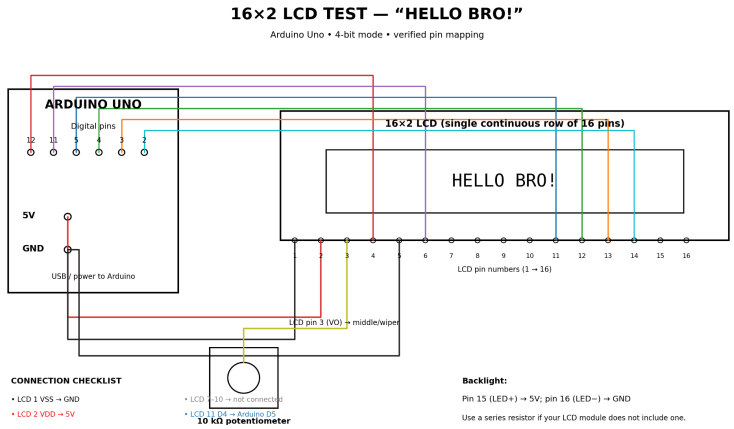
void setup() {
  lcd.begin(16, 2);
  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("HELLO BRO!");

  lcd.setCursor(0, 1);
  lcd.print("LCD WORKING");
}

void loop() {
}
```





Troubleshooting:

The circuit was removed after this for the assembly programming assignment. When we tried connecting the circuit again it didn't work. We had to keep changing the position on the LCD and finally we got a new breadboard on which it worked. Also we got very high temperature around 2crore celsius. This was because one of the jumper wires from PT100 was damaged so the

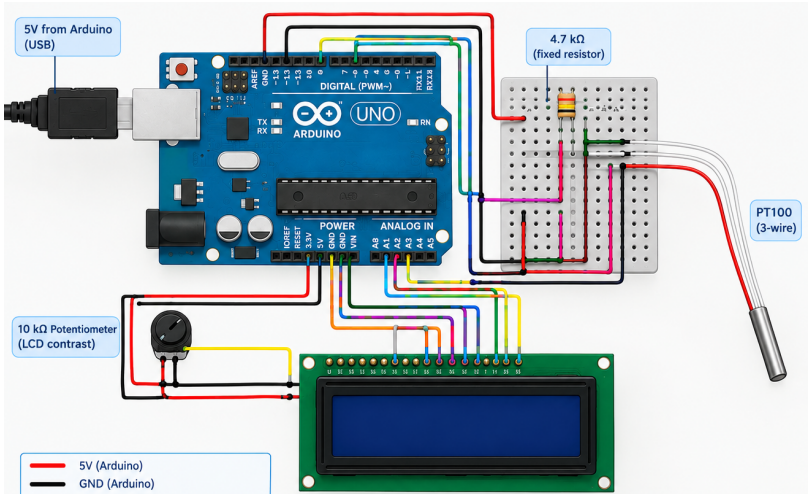
resistance was very high

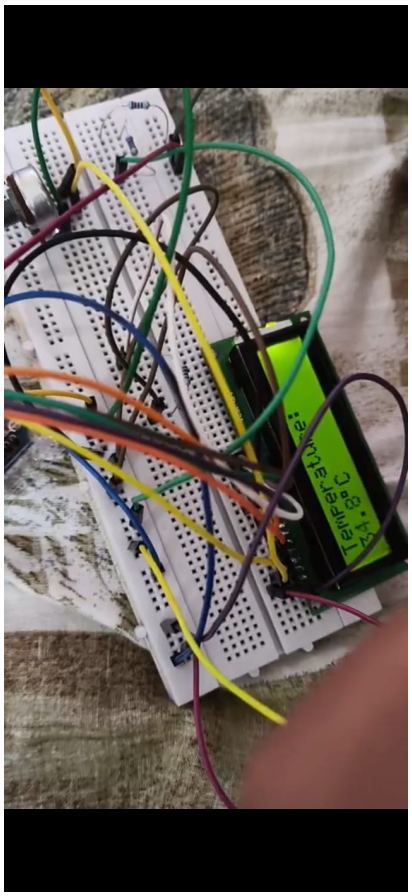


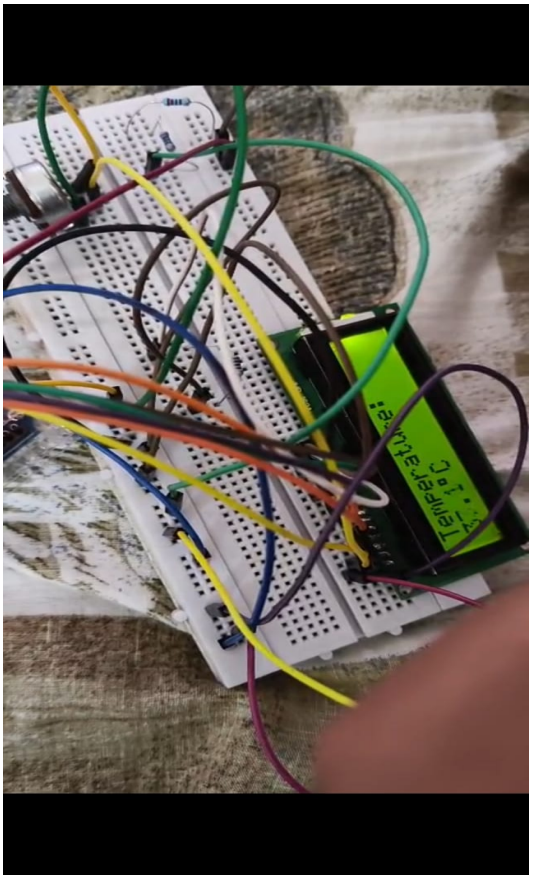
Actual circuit

We had 2200, 200, 100 Ω resistors. For the voltage divider, we searched up which one to use. While the 200 Ω resistor might be more precise as it is more sensitive to current changes. However at 0°C, the resistance of the PT100 is 100Ω while the other resistor would be 200 ohm so the current through PT100 is very high when compared to the current when a resistor of higher value is used.We then thought of using 2200 ohms but the PT100 is getting hot after sometime. Hence, we decided to go ahead with 4400Ω by placing two 2200Ω resistors in series.Because of soldering and the extra lead, the total resistance when measured using a multimeter turned out to be around 4728. This was also first tried on a breadboard before soldering.

Circuit diagram:


5V from Arduino (USB)
4.7 kΩ (fixed resistor)
PT100 (3-wire)
10 kΩ Potentiometer (LCD contrast)
5V (Arduino)
GND (Arduino)
DIGITAL (PWM~)
ARDUINO
UNO
POWER
ANALOG IN





Uploading code and calibration

After the circuit was built, we insulated the tips of all the wires to avoid accidental touching.

First we uploaded the following code to calculate the raw temperature values that are the values measured by the PT100

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
const int PT100_PIN = A0;
const float R_FIXED = 4729.0;
const float R0 = 100.0;
const float ALPHA = 0.00385;

void setup() {
  lcd.begin(16, 2);
  Serial.begin(9600);

  lcd.setCursor(0, 0);
  lcd.print("PT100 Thermometer");
  delay(1500);
  lcd.clear();
}
void loop() {
  static unsigned long lastUpdate = 0;

  if (millis() - lastUpdate < 1000) {
    return;
  }

   lastUpdate = millis();

  long total = 0;

  for (int i = 0; i < 20; i++) {
    total += analogRead(PT100_PIN);
    delay(5);
```

```
 }

 float adcValue = total / 20.0;

 Serial.print("ADC: ");
 Serial.println(adcValue);


  if (adcValue <= 0 || adcValue >= 1023) {
   lcd.setCursor(0, 0);
   lcd.print("Check PT100     ");
   lcd.setCursor(0, 1);
   lcd.print("Wiring / Sensor ");
   return;
 }

 float resistance = R_FIXED * adcValue / (1023.0 - adcValue);
 float temperature = (resistance / R0 - 1.0) / ALPHA;

 lcd.setCursor(0, 0);
 lcd.print("Temperature:    ");

 lcd.setCursor(0, 1);
 lcd.print(temperature, 1);
 lcd.write((byte)223);
 lcd.print("C            ");

 Serial.print("Resistance: ");
 Serial.print(resistance, 2);
 Serial.print(" ohm | Temperature: ");
 Serial.print(temperature, 2);
 Serial.println(" C");
}
```

The values of temperatures obtained from this code for different bodies once the circuit is built on perfboard are used for calibration. This is to avoid the errors caused due to the change in wires when changed from jumper wires to the wires used on perfboard.These values are compared to the values that are obtained by measuring temperatures using a standard thermometer. These values are then calibrated with the raw values to get an equation for the actual temperature.

Calibration process (LSQ)

The equation is of the form $Ax^2 + Bx + C = Y$. These coefficients are obtained by minimizing the sum of squared differences between the reference temperatures and the predicted temperatures. The resulting calibration equation is incorporated into the Arduino program, and its performance is evaluated by comparing calibrated and reference readings.

The calibration has been done using the resistance values (taken as x) at that particular temperature using the measured temperature (taken as Y) as reference.The temperatures have been measured in the course of boiling of water in a kettle and the lower values from cold water. The calculations have been done in excel. The excel sheet has been attached in the repository.
Values of coefficients:
A=0.01226, B=5.606669, C=-443

Final code after calibration:

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

const int PT100_PIN = A0;
const float R_FIXED = 4729.0;

// Your custom Least Squares Regression (LSQ) Coefficients
// Formula format: Temperature = (A * R^2) + (B * R) + C
const float LSQ_A = -0.01226;
const float LSQ_B = 5.606669;
const float LSQ_C = -443.678;

void setup() {
```

```
  lcd.begin(16, 2);
  Serial.begin(9600);

  lcd.setCursor(0, 0);
  lcd.print("PT100 Calibrated");
  delay(1500);
  lcd.clear();
}

void loop() {
  static unsigned long lastUpdate = 0;

  if (millis() - lastUpdate < 1000) {
    return;
  }
  lastUpdate = millis();

  long total = 0;

  // 20-sample averaging over 100ms to eliminate electrical noise
  for (int i = 0; i < 20; i++) {
    total += analogRead(PT100_PIN);
    delay(5);
  }

  float adcValue = total / 20.0;

  Serial.print("ADC: ");
  Serial.println(adcValue);

  if (adcValue <= 0 || adcValue >= 1023) {
    lcd.setCursor(0, 0);
    lcd.print("Check PT100     ");
    lcd.setCursor(0, 1);
    lcd.print("Wiring / Sensor ");
    return;
  }

  // Calculate the physical resistance of the PT100 sensor
  float resistance = R_FIXED * adcValue / (1023.0 - adcValue);

  // Custom LSQ quadratic formula applied here for maximum accuracy
  float temperature = (LSQ_A * resistance * resistance) + (LSQ_B * resistance) + LSQ_C;

  lcd.setCursor(0, 0);
  lcd.print("Temperature:    ");

  lcd.setCursor(0, 1);
  lcd.print(temperature, 1);
```

```
  lcd.write((byte)223); // Degree symbol
  lcd.print("C             ");

  Serial.print("Resistance: ");
  Serial.print(resistance, 2);
  Serial.print(" ohm | Temperature: ");
  Serial.print(temperature, 2);
  Serial.println(" C");
}
```

3D printing

3D modelling was done after finishing the circuit soldering and simultaneously with calibration. In the 3D model, we have two files, the box containing all the components and a lid that can slide on to the box. In the box, there are 3 holes, one for the usb to supply power, one for the PT100 to stick out, and one for the header pins so that the LCD can be attached from the front.The stl and the scad files are in the git repository.